

1. Bacterial Meningitis

Definition

Inflammation of the arachnoid, pia mater, subarachnoid space, and CSF. A life-threatening disorder most often caused by bacteria or viruses. Bacteria enter the CNS via the bloodstream, penetrating trauma, invasive procedures, or ruptured abscess.

Causes

- Bacteria — most serious (N. meningitidis, S. pneumoniae, H. influenzae)
- Viruses, Fungi, Parasites (less common)
- Risk factors: immunosuppression, extremes of age, crowded living, VP shunt, alcohol abuse, under-vaccination

Clinical Manifestations

- Severe headache, high fever & chills
- Nuchal rigidity (stiff neck)
- Nausea / vomiting, photophobia, irritability
- Positive Kernig's sign & Brudzinski's sign
- Signs of increased ICP, altered mental status
- Purpuric rash (meningococcal), seizures

Complications

- SIADH (syndrome of inappropriate ADH)
- DIC — disseminated intravascular coagulation
- Septic shock, Waterhouse-Friderichsen syndrome
- Hydrocephalus, hearing loss
- Permanent neurological deficits

Nursing Diagnoses

- **Risk for Ineffective Cerebral Tissue Perfusion** r/t cerebral edema and increased ICP
- **Hyperthermia** r/t infectious process
- **Acute Pain** r/t meningeal irritation and photophobia
- **Risk for Injury** r/t seizure activity and altered LOC
- **Disturbed Sensory Perception** r/t neurological dysfunction

Nursing Interventions

- Neurological assessment every 1–2 hours (LOC, pupils, motor)
- Maintain quiet, dark environment — reduce stimulation
- HOB elevated 30°, neutral head position
- IV antibiotics STAT — empirical coverage while awaiting CSF results
- Dexamethasone to reduce cerebral inflammation
- Fever management: acetaminophen, cooling blanket
- Droplet isolation precautions for first 24 hrs of antibiotics
- Seizure precautions: padded rails, O₂ at bedside
- Accurate I&O — monitor for SIADH (fluid restriction) vs DI
- Monitor for Cushing's triad: widening pulse pressure, bradycardia, irregular respirations
- ICP monitoring if indicated
- Lumbar puncture findings: ↑ WBC, ↑ protein, ↓ glucose, cloudy CSF

2. Craniotomy

Indications

- Tumor resection (primary or metastatic)
- Cerebral decompression — relieve ICP
- Evacuation of hematoma or abscess
- Clipping or removal of aneurysm
- AVM (arteriovenous malformation) treatment

Complications

- Cerebral edema — peaks 48–72 hrs postop
- Hemorrhage — watch for new neuro deficits
- CSF leak — clear drainage from wound/nose (glucose ≥ 30 mg/dL)
- Diabetes Insipidus — polyuria >200 mL/hr, SG ≤ 1.005
- Infection / meningitis — sterile technique essential
- DVT — from prolonged immobility
- SIADH — fluid & sodium imbalance

Nursing Diagnoses

- **Decreased Intracranial Adaptive Capacity** r/t failure of normal intracranial compensatory mechanisms
- **Ineffective Cerebral Tissue Perfusion** r/t cerebral edema and increased ICP
- **Acute Pain** r/t surgical incision and ICP elevation
- **Risk for Infection** r/t surgical wound and invasive monitoring devices
- **Disturbed Body Image** r/t change in body structure and appearance

Patient Teaching

- Avoid bending, lifting, straining, Valsalva maneuver
- No coughing, sneezing, or nose-blowing postoperatively
- Report clear drainage from wound — may indicate CSF leak
- Report vision changes, seizures, weakness immediately
- Activity restrictions — gradual return to normal
- Wound care and dressing changes
- Address fear of permanent neurological impairment

Nursing Interventions

- HOB 30–45° at all times — facilitates venous drainage
- Maintain neutral head position — avoid neck/hip flexion
- Hourly neurological assessment (LOC, pupils, motor strength)
- Hourly I&O — identify fluid imbalance early
- Measure urine specific gravity if DI suspected
- Pain: small IV morphine (ICU) → acetaminophen + codeine (oral)
- Stool softeners — prevent straining which raises ICP & BP
- Pulmonary care — space activities to prevent ICP spikes
- Sterile technique for incision and drainage device care
- Corneal protection: saline drops Q2h, cover eyes if comatose
- Pad side rails — injury prevention during altered mentation
- Fluid restriction if ordered (cerebral edema or SIADH)
- Monitor for signs of meningitis — fever, nuchal rigidity

3. Diabetic Ketoacidosis (DKA)

Definition

Life-threatening hyperglycemic crisis in DM. Absent insulin → cells use fat for energy → ketone production → metabolic acidosis. The body converts glycogen (glycogenolysis) and protein to glucose, worsening hyperglycemia.

Causes

- Infection — most common precipitating factor
- Omission or inadequate insulin dose
- New-onset undiagnosed Type 1 DM
- Surgery, trauma, or major physical stress
- Psychological stress + eating disorders in young DM patients
- Increased metabolic demand (changes in insulin type/dosage)

Clinical Manifestations

- Polyuria, polydipsia, polyphagia, malaise, headache
- Kussmaul respirations — deep and rapid (air hunger)
- Fruity/acetone breath odor
- Dehydration: dry skin, sunken eyes, hypotension, tachycardia
- Nausea/vomiting, extreme fatigue, weight loss
- CNS depression → confusion → coma if untreated

Investigations

- Blood glucose > 250 mg/dL
- Urine ketones — strongly positive
- ABG: pH < 7.3, ↓ bicarbonate → metabolic acidosis
- Low serum sodium (water shift from ICF to ECF)
- Elevated serum potassium initially (acidosis drives K⁺ out of cells)
- Elevated BUN / creatinine
- CBC, electrolytes, renal/hepatic function

Nursing Diagnoses

- **Deficient Fluid Volume** r/t osmotic diuresis and vomiting
- **Imbalanced Nutrition: Less than Body Requirements** r/t insulin deficiency and cellular starvation
- **Decreased Cardiac Output** r/t alterations in preload from fluid deficit
- **Risk for Electrolyte Imbalance** r/t osmotic diuresis and insulin therapy
- **Anxiety** r/t threat to biologic, psychologic, and social integrity

Nursing Management

- IV fluids: 0.9% NaCl (isotonic) first → 0.45% NaCl if hypernatremia
- Add dextrose when glucose reaches 200 mg/dL to prevent hypoglycemia
- Regular insulin: IV bolus 0.1 unit/kg → infusion 0.1 unit/kg/hr
- Target glucose drop: 50–70 mg/dL/hour
- Monitor blood glucose EVERY HOUR
- Potassium replacement: must be ≥3.3 mEq BEFORE starting insulin
- Add 20–40 mEq/L K⁺ to IV fluids based on serum level
- Bicarbonate: only if pH < 7.10 — give by slow IV infusion
- NG tube if gastric distention, ileus, or vomiting
- NPO during acute phase — advance diet as tolerated
- Monitor for complications: fluid overload, hypoglycemia, cerebral edema
- Transition to SC insulin when glucose 200 mg/dL + acidosis corrected
- Patient education: sick-day rules, insulin management, glucose monitoring

4. Disseminated Intravascular Coagulation (DIC)

Definition

Condition where blood clots form throughout the body, blocking small blood vessels. Micro-clots impair tissue perfusion in major organs → hypoxia, ischemia, tissue damage. Clotting factors are consumed → simultaneous bleeding. Always secondary to an underlying disorder.

Causes

- Infection — bacterial (Gram+/-), viral, fungal, protozoan
- Cancer — especially acute promyelocytic leukemia (APL)
- Obstetric — abruptio placentae, eclampsia, postpartum hemorrhage
- Massive tissue injury — trauma, burns, hyperthermia, surgery
- Transfusion reaction — ABO blood group incompatibility
- Septic shock (most common ICU trigger)

Clinical Manifestations

- Bleeding from multiple body sites simultaneously
- Purpura, petechiae, hematomas, ecchymosis
- Prolonged oozing from IV sites — FIRST sign
- Hypotension, tachycardia, signs of shock
- Hematuria, hemoptysis, GI bleeding
- Blood clots: DVT, pulmonary embolism
- Confusion, fever, oliguria, cyanosis

Investigations

- PT > 15 seconds (prolonged)
- PTT > 60–90 seconds (prolonged)
- Fibrinogen — low (consumed)
- D-dimer / FSPs — HIGH (fibrin split products)
- Platelet count — falling / thrombocytopenia
- ISTH score ≥ 5 = DIC confirmed

Nursing Diagnoses

- **Ineffective Tissue Perfusion** r/t microthrombi formation and hemorrhage
- **Risk for Bleeding** r/t depletion of clotting factors and platelets
- **Impaired Gas Exchange** r/t hemorrhage and microemboli in pulmonary vasculature
- **Acute Pain** r/t tissue ischemia from microthrombi
- **Fear** r/t life-threatening condition and intensive interventions

Nursing Interventions

- Treat underlying cause FIRST (antibiotics for infection, evacuate uterus in obstetric DIC)
- Assess LOC every 1–2 hrs — restlessness/irritability = early cerebral hypoxia
- Monitor O₂ saturation — keep ≥ 90%; monitor ABGs
- Position: high-Fowler's — promotes lung expansion
- Reposition every 2 hrs + chest physiotherapy
- Monitor HR and BP — tachycardia + hypotension = ↓ cardiac output
- Inspect skin for petechiae, purpura, oozing from IV sites
- Administer blood products: RBCs, FFP, platelets, cryoprecipitate
- Monitor coagulation labs: PT, PTT, fibrinogen, D-dimer

Bleeding Precautions

- Avoid IM injections and unnecessary venipunctures
- Use oral or SC medication routes instead
- Gentle oral care — saline rinses instead of toothbrushes
- Apply firm pressure ≥ 5 min to all venipuncture sites
- Prevent falls — padded side rails, non-slip footwear
- Avoid medications that impair platelet function (NSAIDs, aspirin)

5. Hepatic Encephalopathy

Definition

Life-threatening complication of liver failure. The neuropsychiatric manifestation of hepatic failure associated with portal hypertension and shunting of portal blood (containing ammonia) directly into systemic circulation — bypassing liver detoxification.

Causes

- Hepatic insufficiency — liver cannot detoxify metabolic byproducts
- Portal-systemic shunting — collateral vessels bypass liver
- GI bleeding — esophageal varices, chronic GI bleed
- High-protein diet → excess ammonia production
- Bacterial infection / uremia
- Alkalosis or hypokalemia — increases ammonia absorption
- Medications: sedatives, tranquilizers, analgesics, diuretics
- Excessive diuresis, dehydration, surgery, fever

Stages

- Stage 1: Slight confusion, mood/sleep changes, sleep during day (insomnia at night)
- Stage 2: Drowsy, asterix (hand flapping), disorientation to time
- Stage 3: Difficult to awaken, complete disorientation, marked confusion
- Stage 4: Coma, possible seizures, no response to stimuli

Clinical Manifestations

- Mental status changes — from mild confusion to coma
- Sleep-wake cycle reversal
- Asterix — flapping tremor of outstretched hands
- Progressive disorientation, seizures
- EEG: generalized slowing, triphasic waves
- ↑ Serum ammonia level
- Deteriorating handwriting and arithmetic scores

Nursing Interventions

- Lactulose: decreases colonic pH → reduces ammonia absorption
- Neomycin / Rifaximin: destroy bowel bacteria → reduce ammonia production
- Diet: protein 1.0–1.5 g/kg/day (do NOT over-restrict)
- Neurological assessment frequently — record handwriting & arithmetic daily
- Monitor mental status (LOC, behavior, orientation)
- Accurate daily I&O, body weight, vital signs Q4h
- Safe environment: prevent injury, bleeding, infection
- Encourage deep breathing + position changes — prevent pneumonia
- Assess for potential infection sites (peritoneum, lungs) — report promptly

Patient Teaching

- Family: watch for subtle early signs of recurrence after discharge — mild confusion, personality changes, sleep disturbance
- Diet: moderate-protein, high-calorie (protein 1.0–1.5 g/kg)
- In acute phase: reduce protein to 0.8 g/kg briefly
- Add protein back gradually: 10 g increments every 3–5 days if mental status stable
- Never abruptly restrict protein long-term — causes muscle wasting
- Avoid alcohol and all hepatotoxic drugs/supplements
- Take lactulose as prescribed (target: 2–4 soft stools/day)
- Regular follow-up appointments and lab monitoring
- Report confusion, personality changes, increased drowsiness immediately

6. Hypertensive Crisis

Definition

Severe elevation of BP > 180/110 mmHg that can lead to stroke and end-organ damage. Classified into two types based on presence or absence of acute target organ damage.

Types: Urgency vs Emergency

- **URGENCY:** Severe BP elevation — NO acute organ damage. Treated with oral antihypertensives. Can be managed in outpatient or ED setting.
- **EMERGENCY:** Severe BP elevation + acute end-organ damage (brain, heart, kidneys). Requires IV treatment in ICU. Life-threatening.

Causes

- Primary (essential) hypertension — most common
- Abrupt withdrawal of antihypertensive drugs
- Pheochromocytoma (catecholamine surge)
- Eclampsia / pregnancy-induced hypertension
- Renal artery stenosis / renovascular disease
- Myocardial ischemia, aortic dissection
- Stimulant drugs (cocaine, amphetamines)

Clinical Manifestations

Emergency:

- Severe headache, altered LOC, papilledema
- Chest pain / angina, pulmonary edema, dyspnea
- Oliguria, hematuria — renal involvement
- Neurological deficits, visual changes

Urgency:

- Severe headache, epistaxis (nosebleeds)
- Severe anxiety, shortness of breath — NO organ damage

Complications

- Stroke, hypertensive encephalopathy
- Myocardial infarction (MI), aortic dissection
- Acute renal failure, pulmonary edema
- Retinal damage, blindness

Nursing Diagnoses

- **Decreased Cardiac Output** r/t increased systemic vascular resistance
- **Risk for Ineffective Cerebral Tissue Perfusion** r/t hypertensive encephalopathy and CVA
- **Excess Fluid Volume** r/t sodium and water retention
- **Anxiety** r/t change in health status and threat to biologic integrity
- **Deficient Knowledge** r/t disease process and antihypertensive regimen

Nursing Interventions

- Continuous hemodynamic monitoring — arterial line preferred
- IV antihypertensives: nitroprusside, labetalol, nicardipine
- BP reduction target: ↓25% in first hour → 160/100 over next 6 hrs → gradual over days
- Exception — aortic dissection: target systolic < 100 mmHg if tolerated
- Neurological assessment Q1h — monitor for stroke signs
- Urinary catheter — monitor output Q1h (renal perfusion)
- Continuous cardiac monitoring (ECG)
- Seizure precautions, bed rest in quiet environment
- CBC, electrolytes, BUN, creatinine, urinalysis, chest X-ray, ECG

Patient Education

- Take antihypertensive medications DAILY — never stop abruptly
- Low-sodium diet (avoid canned foods, processed meats, soy sauce)
- Weight loss + regular physical activity 30 min/day
- Limit alcohol, stop smoking
- Regular home BP monitoring — know target values
- Stress management: yoga, relaxation techniques
- Report severe headache, visual changes, or chest pain immediately

- Follow-up: Q1–2 weeks until target achieved, then Q6 months

7. Increased Intracranial Pressure (ICP)

Definition

Sustained increase in pressure within the rigid cranial vault > 20 mmHg. The vault contains 3 non-compressible components: brain tissue, blood, and CSF. If one increases, others must decrease (Monro-Kellie doctrine) or ICP rises. Normal ICP: 0–15 mmHg. CPP = MAP – ICP. Goal: CPP 60–70 mmHg.

Causes

- Traumatic brain injury (TBI)
- Stroke (ischemic or hemorrhagic)
- Brain tumor or abscess
- Hydrocephalus
- Cerebral edema (post-surgery, infection)
- Subarachnoid or intracerebral hemorrhage
- Meningitis / encephalitis

Clinical Manifestations — Cushing's Triad

- Widening pulse pressure (systolic ↑, diastolic ↓)
- Bradycardia
- Irregular respirations (Cheyne-Stokes)

Other signs:

- ↓ LOC — EARLIEST and most sensitive sign
- Pupillary changes: unequal, sluggish, or fixed & dilated
- Vomiting without nausea (projectile)
- Papilledema (blurred disc margins on fundoscopy)
- Morning headache (worse lying down)
- Decorticate → decerebrate posturing (late, severe)

Investigations

- CT scan — first-line (identify blood, edema, tumor)
- MRI — more detailed structural imaging
- ICP monitoring: EVD (gold standard), fiberoptic probe
- ABGs — monitor for hypoxia (PaO₂) and CO₂ (ventilation)
- Lumbar puncture — CONTRAINDICATED if ICP is elevated

Nursing Precautions

- HOB at 30° — neutral head position at all times
- Avoid neck/hip flexion — impairs venous drainage
- Space ALL nursing activities — allow rest between turns/hygiene
- Avoid Valsalva maneuver: no straining, give stool softeners
- Minimize sensory stimulation — noise, painful stimuli, bright light
- Seizure precautions — padded rails, suction at bedside
- Accurate hourly I&O — monitor for DI (polyuria) vs SIADH
- Assess physiologic response to visitors — limit if ICP rises

Medical Management

- Osmotic diuretics: 25% Mannitol or 3% Hypertonic Saline
- Corticosteroids (dexamethasone) — for tumor/abscess-related edema
- Controlled hyperventilation (intubated): PCO₂ target 30–35 mmHg
- CSF drainage via EVD — reduces ICP rapidly
- Sedation and paralysis — reduce metabolic demands
- Temperature management — prevent hyperthermia (worsens edema)
- Decompressive craniectomy — last resort for refractory ICP

- Antipyretics, maintain normoglycemia, correct hyponatremia

8. Liver Dialysis

Definition

A system designed to remove toxins not cleared by the failing liver. Targets lipophilic and albumin-bound substances (bilirubin, bile acids), metabolites of aromatic amino acids, and medium-sized molecules that accumulate in liver failure and cause organ dysfunction.

Types of Systems

- BIOLOGICAL systems: contain functioning liver cells → provide BOTH detoxifying AND synthetic liver functions. More complex and expensive.
- NON-BIOLOGICAL systems: no living cells → provide detoxification ONLY (no synthetic function). Examples: MARS (Molecular Adsorbent Recirculating System), Prometheus. Simpler and cheaper.

Settings Used In

- Transplantation centers
- Intensive Care Units (ICU)
- Specialized dialysis units

Indications

- Acute hepatic encephalopathy from worsening chronic liver disease (acute-on-chronic) or fulminant hepatic failure
- Severe drug overdose — acetaminophen toxicity, tricyclic antidepressants
- Acute decompensation of chronic liver disease
- Acute liver failure awaiting transplantation (bridge to transplant)
- Liver dysfunction following transplantation (primary graft non-function)

Complications

- Thrombocytopenia — platelet destruction during circuit
- Coagulopathy — clotting factor loss
- Hypotension — hemodynamic instability during sessions
- Electrolyte imbalances — monitor Na^+ , K^+ , Ca^{2+} , Mg^{2+}
- Infection risk — from extracorporeal circuit
- Long-term side effects — patients must be counseled before treatment

Clinical Efficiency / Benefits

- Complete recovery of liver function following recompensation (acute-on-chronic failure)
- Bridge short-term or long-term waiting times until liver transplantation
- Increasing the survival rate of liver transplants — better patient condition at time of transplant
- Improving quality of life during liver failure — reduces encephalopathy, pruritus, fatigue
- Reduction in bilirubin, bile acids, ammonia, and other toxin levels
- May improve hemodynamic stability in hepatorenal syndrome

Key Nursing Points

- Monitor vital signs and hemodynamics throughout session
- Assess neurological status — improvement in encephalopathy
- Monitor coagulation labs: PT, PTT, platelet count before and after
- Ensure IV access is adequate — large-bore or central line
- Document fluid balance accurately during treatment
- Patient/family education: purpose, procedure, expected outcomes

9. Liver Transplantation

Definition

Partial or total removal of the diseased liver and its replacement with a healthy donor liver in the same anatomic location (orthotopic). Used to treat life-threatening, end-stage liver disease for which no other treatment is available.

Indications

- End-stage liver disease (cirrhosis) from any cause
- Fulminant hepatic failure — acute liver failure
- Hepatocellular carcinoma (within Milan criteria)
- Biliary atresia (common indication in children)
- Metabolic liver diseases (Wilson's disease, α_1 -antitrypsin deficiency)
- Primary sclerosing cholangitis, primary biliary cirrhosis

Contraindications

- Sepsis — active systemic infection
- Malignancies outside the liver
- Active substance abuse (alcohol, drugs)
- Irreversible brain damage
- Inability to comply with post-transplant immunosuppression
- AIDS (active), active cardiopulmonary disease

Types of Rejection

- **HYPERACUTE:** Within minutes to hours (extremely rare in liver). Antibody-mediated. Usually leads to immediate graft loss.
- **ACUTE (most common):** Days to weeks post-transplant. Cell-mediated. Treated with high-dose IV corticosteroids. Most cases respond well.
- **CHRONIC:** Months to years post-transplant. Progressive bile duct damage. Leads to graft failure. May require re-transplantation.

Nursing Diagnoses

- **Risk for Infection** r/t immunosuppressive therapy and surgical wound
- **Risk for Bleeding** r/t coagulopathy and surgical trauma
- **Impaired Fluid Volume** r/t third spacing, hemorrhage, and low oncotic pressure
- **Anxiety** r/t uncertainty of graft function and complex postoperative care
- **Deficient Knowledge** r/t immunosuppression regimen and rejection signs

Nursing Care

- Hemodynamic monitoring: watch for hypotension during warming phase, hypertension from pain/stress
- Fluid resuscitation: crystalloids (NS 0.9%), albumin, blood products as ordered
- Monitor renal function closely — nephrotoxic immunosuppressants: cyclosporine, tacrolimus
- Watch for infection signs — immunosuppression increases risk significantly
- Wound care — sterile technique; inspect incision for dehiscence or infection
- Accurate I&O — monitor for oliguria (hepatorenal syndrome may persist)
- Administer immunosuppressants on time — tacrolimus, mycophenolate, steroids
- Assess for rejection: fever, jaundice, right upper quadrant pain, elevated LFTs
- Manage ascites: sodium restriction, diuretics (spironolactone + furosemide)
- Monitor coagulation: PT, PTT — correct with FFP, Vitamin K
- Patient education: rejection signs, infection prevention, medication compliance

10. Nutritional Support in Critical Care

Enteral Nutrition (EN) — Definition

Nutrition delivered via the GI tract (tube feeding). Routes: nasogastric (NGT), nasoduodenal, jejunostomy (PEG/PEJ). Preferred over TPN when GI tract is functional.

EN — Indications

- Functioning GI tract with inability to eat orally
- Burns, trauma, head injury, sepsis
- Post-surgery when GI function returns
- Neurological impairment affecting swallowing

EN — Complications

- Aspiration pneumonia — most dangerous (keep HOB 30–45°)
- Diarrhea — most common complication in ICU
- Tube displacement or blockage
- Abdominal distension, constipation
- Refeeding syndrome — hypophosphatemia, hypokalemia
- Sinusitis (nasogastric route)

TPN — Definition

Delivery of ALL nutrients by IV infusion, completely bypassing the GI tract. Recommended when oral or tube feeding is contraindicated or inadequate for more than 5–7 days.

TPN — Indications

- Non-functioning GI tract / prolonged ileus
- Short bowel syndrome, severe malabsorption
- GI fistula, bowel obstruction
- Severe acute pancreatitis (initially)
- Severe inflammatory bowel disease

TPN — Complications

- Sepsis / infection — most serious (strict asepsis required)
- Hyperglycemia — most common metabolic complication
- Hypoglycemia if stopped abruptly
- Electrolyte imbalances — Na⁺, K⁺, Mg²⁺, Phosphate
- Refeeding syndrome
- Liver dysfunction (cholestasis) with long-term use
- Pneumothorax — from central line insertion
- Thrombosis at catheter site

Nursing Care

- Accurately record all food and fluids consumed (type + amount)
- Calculate daily protein and calorie intake
- Anthropometric measurements: height, weight, BMI, triceps skinfold, mid-arm circumference

Enteral Feeding Nursing Care:

- Verify tube placement before each feeding (X-ray for initial; pH / aspiration for ongoing)
- Keep HOB elevated 30–45° continuously during feeding
- Check gastric residual volume every 4–6 hrs — hold if >250 mL
- Monitor bowel sounds and abdominal distension

TPN Nursing Care:

- Use volumetric infusion pump — precise rate control
- Change TPN bag and tubing every 24 hours (strict aseptic technique)
- Monitor blood glucose every 4–6 hrs — treat hyperglycemia with insulin
- Monitor electrolytes, albumin, pre-albumin daily initially
- Inspect central line insertion site for signs of infection
- NEVER stop TPN abruptly — taper slowly to prevent rebound hypoglycemia
- Nursing diagnoses: Imbalanced Nutrition r/t malabsorption; Risk for Infection r/t IV access

11. Renal Transplantation

Contraindications

ABSOLUTE:

- Age < 18 years (living donor)
- Uncontrolled hypertension
- Diabetes mellitus (for living donor)
- Proteinuria > 300 mg / 24 hours
- Abnormal GFR for age
- Microscopic hematuria
- High risk of thromboembolism
- Significant chronic illness: lung disease, recent malignancy, heart disease
- History of bilateral kidney stones
- HIV positive

RELATIVE:

- Anatomical anomalies of the graft
- Active chronic infection (TB, Hepatitis B/C, parasites)
- Obesity, psychiatric disorders

Nursing Care

Immediate Postoperative:

- Maintain fluid & electrolyte balance — strict I&O
- Wound care — sterile dressing changes
- Pain management around the clock
- Pulmonary hygiene: incentive spirometry + early ambulation
- Restore normal bowel function (ileus risk from steroids)

Long-term:

- Lifelong follow-up by transplant specialist Q6–12 months
- Renal function + immunosuppression monitoring Q4–8 weeks
- Annual dermatological exam, tumor screening, chest X-ray
- Monitor proteinuria, BP, lipids, glucose, bone density

Complications

- Infection — leading cause of morbidity (immunosuppressed)
- CHF — most common cause of hospitalization
- Rejection (hyperacute, acute, chronic)
- Hypertension, renal failure
- Metabolic complications: hyperglycemia, dyslipidemia

Patient Teaching

Call your doctor immediately if:

- Fever > 38°C (100°F)
- Flu-like symptoms: chills, aches, dizziness, nausea/vomiting
- New pain or tenderness around transplanted kidney
- Fluid retention / swelling
- Weight gain > 1–2 kg within 24 hours
- Significant decrease in urine output

Lifestyle:

- Exercise: 20–30 min, 3–4 times/week
- Wait 1.5 hrs after eating before exercising
- Avoid heavy lifting, pushing, or strenuous exercise initially
- Wear protective footwear, dress appropriately

Medication:

- Take immunosuppressants EXACTLY as prescribed — NEVER skip doses
- Avoid NSAIDs and nephrotoxic agents
- Take beta-blockers as prescribed — do NOT stop abruptly
- Sun protection — increased skin cancer risk
- Follow recommended diet: low sodium, low fat if needed

12. Seizures

Definition

Repetitive depolarization of hyperactive and hypersensitive brain cells → abnormal electrical activity throughout the brain. Uncontrolled discharge propagates through neural pathways, producing clinical manifestations (motor, sensory, autonomic, behavioral).

Causes

- Epilepsy — most common chronic cause
- Drug or alcohol abuse / withdrawal
- Drug toxicity — aminophylline, lidocaine, isoniazid
- Recent head injury / TBI
- CNS infection — meningitis, encephalitis
- CNS tumor, abscess
- Acute metabolic disturbances: hypoglycemia, hyponatremia, hypocalcemia, renal failure
- Cerebrovascular accident (CVA)
- Hypoxemia, high fever

Nursing Care DURING Seizure

- PRIMARY goal: Protect airway and prevent injury
- Position: side-lying — facilitates airway clearance
- Do NOT restrain the patient — guide movements gently
- Do NOT force an airway into the mouth — risk of dental/jaw injury
- Clear area of hard/sharp objects
- Observe and document: seizure type, point of origin, spread, duration
- Note: automatisms (lip smacking, repeated swallowing), bowel/bladder incontinence, tongue biting
- Assess LOC before, during, and after
- Administer O₂ via mask + suction as needed

Medications:

- Lorazepam (Ativan): 0.1 mg/kg IV — up to 4 mg; max 2 mg/min; may repeat in 5–10 min
- Midazolam: 0.2 mg/kg IM — up to 10 mg max
- Diazepam (Valium): 0.15 mg/kg IV — max 10 mg; max 5 mg/min
- Phenytoin (Dilantin): 20 mg/kg IV — max 50 mg/min (long-acting)

Nursing Care POSTICTAL

- Assess vital signs closely — provide ECG monitoring
- Monitor oxygenation: ABGs, SpO₂, breath sounds
- Maintain side-lying position — aspiration risk
- Check orientation, ability to speak (patient usually sleeps afterward)
- Note: headache, myalgia, fatigue, confusion (normal postictal features)
- Document seizure characteristics completely: onset, duration, type, spread, recovery time
- Obtain: EEG, CT/MRI, serum drug screen, electrolytes (BUN, Ca²⁺, Mg²⁺, glucose), CBC
- Provide emotional support — seizures are frightening and disorienting
- Reorient patient calmly as consciousness returns
- Continue seizure precautions: padded rails, O₂ at bedside, suction available

Key Assessment Questions

- Identify precipitating factors and preceding aura
- What was patient doing at the time of seizure?
- Were there unusual symptoms before (movements, sensations, smells)?
- Did patient know a seizure was coming?
- History of prior seizures, medications, compliance?

13. Stroke ☆☆☆

Definition & Types

Sudden onset of acute neurological deficit persisting > 24 hours, caused by interruption of blood flow to the brain. TIA: same but resolves in < 24 hrs (usually < 1 hr).

- ISCHEMIC — 87%: Thrombotic (75%) | Embolic (25%)
- HEMORRHAGIC — 13%: Intracerebral (10%) | Subarachnoid (3%)

Causes

- Ischemic: vascular obstruction (thrombus or embolus), small penetrating artery thrombosis, cardiac embolism (atrial fibrillation)
- Hemorrhagic: hypertensive vascular disease, ruptured aneurysm (weakness in artery wall), ruptured AVM (common in young people)

Risk Factors

- Hypertension — most important modifiable risk factor
- Atrial fibrillation, DM, hyperlipidemia, obesity, smoking
- Age, family history, prior TIA or stroke

Clinical Manifestations

- Numbness/weakness of face, arm, or leg (one side)
- Confusion, trouble speaking (dysarthria) or understanding (aphasia)
- Visual: diplopia, homonymous hemianopsia, loss of peripheral vision
- Hemiparesis/hemiplegia, ataxia, dysphagia

Investigations

- CT without contrast — FIRST-LINE: differentiates hemorrhagic from ischemic
- MRI — detects infarction earlier; determines type, size, location
- CT/MRI Angiography — confirms aneurysm or AVM
- 12-lead ECG + continuous cardiac monitoring
- Echocardiography (for cardioembolic source)
- CBC, electrolytes, glucose, renal/hepatic function, coagulation
- ABGs if hypoxia suspected, Transcranial Doppler

rtPA (Thrombolytic Therapy)

For ISCHEMIC stroke ONLY:

- Dose: 0.9 mg/kg (max 90 mg)
- 10% as IV bolus over 1–2 min; 90% infusion over 60 min
- Window: ≤ 4.5 hours from symptom onset
- No antithrombotic therapy for 24 hours after rtPA
- Risk: intracerebral hemorrhage ~6.4%
- Inclusion: ischemic diagnosis confirmed, SBP ≤185 mmHg, DBP ≤110 mmHg, not minor or rapidly resolving stroke

Nursing Diagnoses

- **Ineffective Cerebral Tissue Perfusion** r/t interruption of cerebral blood flow
- **Impaired Physical Mobility** r/t hemiplegia and neuromuscular impairment
- **Impaired Verbal Communication** r/t aphasia from cerebral damage

Nursing Interventions

- Neurological assessment Q1–2h: LOC, pupils, motor strength, speech, facial droop
- Airway management — priority; O₂ therapy for SpO₂ > 94%
- HOB at 30° — reduces ICP and aspiration risk
- Monitor BP — for ischemic: allow permissive hypertension (do NOT aggressively lower unless >220/120 or before rtPA); for hemorrhagic: target SBP <140 mmHg
- Dysphagia screening before ANY oral intake — bedside swallowing test
- NPO if dysphagia confirmed — consult speech therapy
- Aspiration precautions: HOB up, thickened liquids if ordered
- DVT prophylaxis: compression stockings, sequential compression devices
- Bladder care: intermittent catheterization preferred over indwelling
- Skin integrity: reposition Q2h, pressure injury prevention
- Emotional support: explain aphasia/paralysis to patient AND family
- Early rehabilitation: physiotherapy, OT, speech therapy from day 1
- Anticoagulation for embolic stroke (heparin → warfarin)
- Antiplatelet therapy: aspirin or clopidogrel for thrombotic stroke
- Monitor glucose — maintain normoglycemia (hyperglycemia worsens outcomes)

- Hemorrhagic: 'exploding headache,' decreased LOC, nuchal rigidity
- Emotional lability, loss of self-control

- **Risk for Aspiration** r/t dysphagia and reduced LOC

14. Thyroid Storm (Thyrotoxic Crisis)

Definition

Severe, life-threatening form of hyperthyroidism with abrupt sudden onset. Represents extreme hypermetabolic state from excessive thyroid hormone. Almost always fatal if untreated. Requires ICU-level care.

Causes (Precipitating Triggers)

In a patient with pre-existing hyperthyroidism:

- Infection — most common trigger
- Surgery or trauma — any major surgical procedure
- Pregnancy or delivery
- Abrupt withdrawal of antithyroid medications
- Amiodarone — causes thyroid dysfunction in 10–20% of patients
- Major physical or psychological stress

Clinical Manifestations

Cardiac:

- Tachycardia, atrial fibrillation/PVCs, palpitations
- Hypertension, chest pain, acute heart failure, pulmonary edema

Neurological:

- Hyperkinesia, nervousness, muscle weakness
- Confusion, convulsions, fine tremor, psychosis, diaphoresis

GI:

- N/V, diarrhea, hepatomegaly, abdominal pain, weight loss

Thermoregulatory:

- High fever (hyperthermia), profuse sweating

Investigations

- Free T3 and T4 — significantly elevated
- TSH — suppressed or undetectable
- CBC, liver enzymes, glucose, electrolytes
- Cortisol level (stress response)
- ECG — tachyarrhythmias, atrial fibrillation

Nursing Interventions

Medication Sequence (critical order):

- 1. Beta-blockers (propranolol): FIRST — blocks cardiovascular effects of excess hormone
- 2. PTU (propylthiouracil): SECOND — blocks NEW hormone synthesis
- 3. Iodine (Lugol's): 1–2 hrs AFTER PTU — blocks hormone RELEASE (if given before PTU, used to make more hormone)
- Acetaminophen for fever — NOT aspirin (aspirin displaces T3/T4 from protein binding → worsens crisis)
- Cooling blanket, ice packs to axilla/groin, tepid sponge bath
- IV fluids: large volumes glucose + isotonic sodium to replace losses
- Insulin for hyperglycemia; antibiotics if infection present
- Calm environment for agitated patient — reassure patient/family
- Monitor heart rate closely — should decrease with beta-blockade
- Frequent temperature monitoring until normal
- Monitor for hyponatremia

Patient Teaching

- Take antithyroid medications EXACTLY as prescribed — NEVER stop suddenly
- Avoid iodine-rich foods: seafood, iodized salt, kelp
- Avoid stimulants: caffeine, decongestants, OTC cold medications
- Report immediately: palpitations, fever, tremor, worsening symptoms
- Regular thyroid function tests (T3, T4, TSH) as scheduled
- Stress management techniques — stress triggers thyroid storm
- For patients on antithyroid drugs: report signs of agranulocytosis — sudden sore throat, fever, cough (stop medication, call doctor)
- For beta-blocker patients: report bradycardia, unexplained fatigue, orthostatic hypotension
- Wear medical alert bracelet identifying thyroid condition
- Regular follow-up appointments — do not miss

Nursing Diagnoses

- **Decreased Cardiac Output** r/t tachyarrhythmias and increased metabolic demands
- **Hyperthermia** r/t hypermetabolic state and excessive heat production
- **Deficient Fluid Volume** r/t profuse diaphoresis, vomiting, and diarrhea

- **Anxiety / Acute Confusion** r/t excess thyroid hormone affecting CNS
- **Deficient Knowledge** r/t disease process and long-term medication management

15. Myasthenia Gravis (MG)

Definition

Chronic autoimmune neuromuscular disorder producing sporadic but progressive weakness and abnormal fatigue in skeletal muscles. Antibodies block, destroy, or weaken ACh receptor sites at the neuromuscular junction → muscles cannot contract properly. Weakness **WORSENS** with exercise/repeated movement, **IMPROVES** with rest.

Clinical Manifestations

- Gradual progressive skeletal muscle weakness worsening throughout the day
- Ptosis — drooping of one or both eyelids (hallmark early sign)
- Diplopia — double vision (extraocular muscle weakness)
- Blank/mask-like facial expression (facial muscle weakness)
- Dysphagia — difficulty swallowing
- Dysarthria — slurred speech, difficulty chewing
- Difficulty with lateral head motion
- Respiratory failure if respiratory muscles involved — **MEDICAL EMERGENCY**

Investigations

- Tensilon Test (neostigmine IV): temporary improvement = positive = MG confirmed. Long-standing ocular dysfunction may not respond.
- EMG — electromyography: differentiates nerve disorders from muscle disorders
- Nerve conduction studies: test for ACh receptor antibodies
- CT chest: detect thymoma (associated in many MG patients)
- Serum anti-AChR antibodies — positive in 85% of MG

Myasthenic Crisis

Emergency: sudden severe respiratory distress due to undermedication or triggers (infection, stress, surgery, extreme temperature).

- Signs: acute respiratory distress, increased muscle weakness, difficulty talking/chewing, tachycardia, tachypnea
- Tx: Tracheotomy, positive-pressure ventilation, vigorous suctioning
- IV neostigmine or pyridostigmine; plasmapheresis to remove antibodies; IVIG
- ⚠ Anticholinesterases NOT effective during crisis — **HOLD** until respiratory function improves

Nursing Interventions

- Establish neurologic and respiratory **BASELINE** — before each shift
- Help remove secretions as they accumulate — suction if needed
- Be **ALERT** for signs of impending crisis: increasing weakness, respiratory distress, difficulty talking
- Administer anticholinesterase drugs at **EVENLY SPACED INTERVALS** on time as ordered — timing is critical
- Be prepared to give **ATROPINE** for anticholinesterase overdose or toxicity
- Plan exercise, meals, and activities at **PEAKS** in patient's energy level
- Provide soft, solid foods (**NOT** liquids) — reduces choking risk
- Always sit patient upright to eat
- Respiratory monitoring: SpO₂, breath sounds, respiratory rate
- Encourage patient to take active role in care decisions

Patient Teaching

- Recognize adverse effects of anticholinesterase drugs: headaches, weakness, sweating, abdominal cramps, N/V/D, excessive salivation, bronchospasm — cholinergic crisis (opposite of MG: too much medication)
- Take medications on time **EVERY DAY** — timing is critical

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| | | <ul style="list-style-type: none">• Plan daily activities at energy peaks; frequent rest periods throughout the day• Avoid: strenuous exercise, stress, infection, extreme temperature, sun exposure• Avoid new medications without consulting physician first• Carry medical alert ID at all times• Report worsening symptoms immediately — do not wait |
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